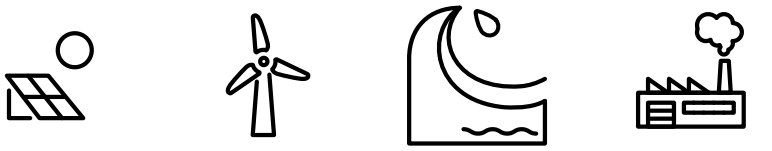
PHYSICAL AND CHEMICAL SCIENCE

Energy Resources: Renewable, Non-renewable and Green Energy

Fossil fuels and renewables, climate change and energy access as global issues, and green energy, with real data



solar **wind** **hydro** **fossil fuels**

In this pack

Learn it: Energy resources: renewable, non-renewable and green ★☆☆
Fact check: Renewable or not? ★☆☆
Think it through: Weighing up energy choices ★★☆
Investigate: Does the number of blades change a wind turbine's speed? ★★☆
Real-world case study: Bangladesh's energy future ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Energy resources: renewable, non-renewable and green

Name: Class: Date:

Learning goals

- Classify energy resources as renewable or non-renewable.
- Explain why burning fossil fuels is a global problem.
- Use real data to compare how the world and Bangladesh make electricity.

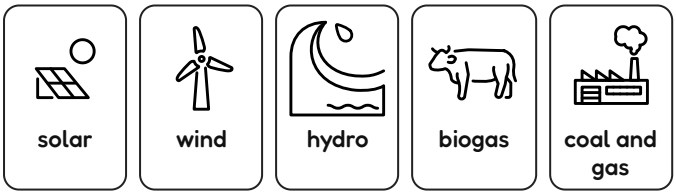
Every time you switch on a fan, the electricity comes from a **power station** using an **energy resource**. Some resources will run out. Some never will. Our choices affect the whole planet.

Non-renewable resources

Fossil fuels: **coal**, **oil** and **natural gas**, formed from plants and animals buried for **millions of years**. We use them much faster than they form, so one day they will **run out**. **Nuclear** fuel (uranium) is also non-renewable, but it does not release carbon dioxide.

Renewable resources

Renewable resources are naturally replaced and will not run out: **solar** (sunlight), **wind**, **hydro** (moving water), **biomass and biogas** (plants and animal dung), **tidal** and **geothermal** (heat from inside the Earth).



Why it is a global issue

- Burning fossil fuels releases **carbon dioxide (CO_2)**, a greenhouse gas that traps heat and causes **climate change**.
- Bangladesh is one of the countries most at risk: stronger cyclones, floods and sea-level rise along the coast.
- Smoke from burning coal, oil and wood also pollutes the air we breathe.
- About **685 million people** in the world still had **no electricity** at home in 2022, mostly in Sub-Saharan Africa.

Green energy

Green (clean) energy comes from renewable resources that release little or no pollution: solar, wind and hydro. Using energy **efficiently** (LED bulbs, switching off) is green too: the cleanest energy is the energy we do not waste.

How the world made its electricity (2024)

Coal	Gas	Other fossil	Nuclear	Hydro	Wind	Solar	Other renewable
34%	22%	3%	9%	14%	8%	7%	3%

About **59%** came from fossil fuels and about **41%** from clean sources (renewables and nuclear). In 2024 wind and solar together made more electricity than hydro for the first time.

How Bangladesh made its electricity (2023)

Natural gas	Oil (liquid fuels)	Coal	Imports, hydro and solar
55%	20%	14%	11%

Bangladesh's own renewables are still small: the **Kaptai** hydro plant on the Karnaphuli River and a growing amount of **solar**, including millions of solar home systems.

Where the numbers come from

Sources (rounded): world, Ember *Global Electricity Review 2025*; Bangladesh, Power Grid Company of Bangladesh data reported by Reuters (2024); people without electricity, *Tracking SDG7: The Energy Progress Report 2024*.

Key words

fossil fuel	coal, oil or gas formed over millions of years
renewable	naturally replaced; will not run out
non-renewable	will run out one day
green energy	clean energy from renewable resources
greenhouse gas	a gas such as CO_2 that traps heat in the air
efficiency	getting the job done with less energy

FACT CHECK · C10 · ENERGY RESOURCES

Renewable or not?

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Classify energy resources.
- Read data from a table.

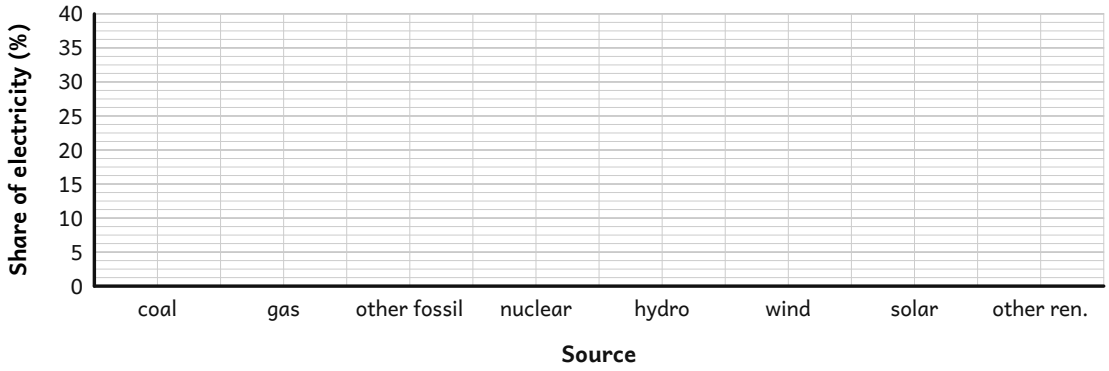
1 Sort the energy resources. [8]

SORT THESE

natural gas **biogas** **wind** **solar** **uranium** **hydro** **oil** **coal**

Renewable	Non-renewable

2 Draw a bar chart of the world's electricity in 2024. [4]



3 Use the chart. [2]

a) Which single source made the most electricity?

.....

b) Add up wind + solar + hydro + other renewables.

.....

THINK IT THROUGH · C10 · ENERGY RESOURCES

Weighing up energy choices

★★☆ Grade 5

Name: Class: Date:

Learning goals

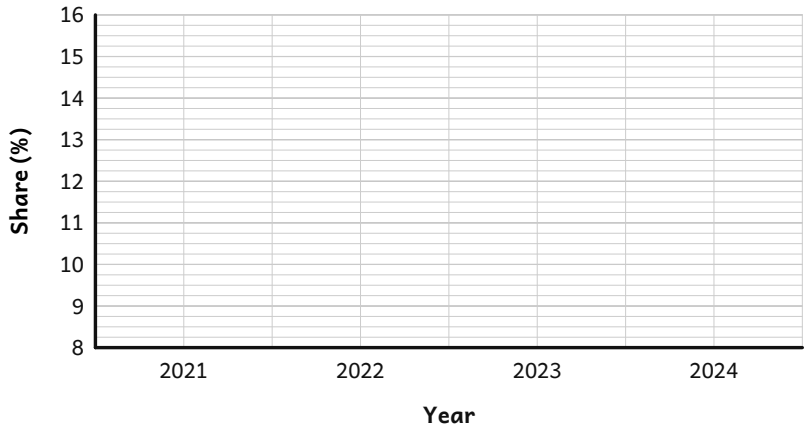
- Compare advantages and disadvantages.
- Interpret a trend.

1 **Complete the table.** [8]

Resource	One advantage	One disadvantage
coal		
solar		
wind		
hydro		

2 **The share of the world's electricity from wind and solar together is shown below. Plot it as a line graph.** [3]

Year	2021	2022	2023	2024
Wind + solar (%)	10.3	11.9	13.4	15.0



3 **Describe the trend. Predict the share in 2025 if it continues.** [2]

.....

.....

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Does the number of blades change a wind turbine's speed?

★★☆ Grade 5

Name: Class: Date:

Learning goals

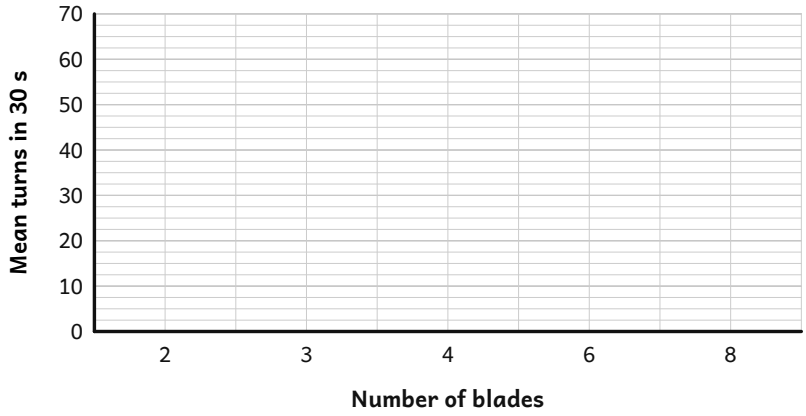
- Plan a fair test with a model.
- Analyse results.

Class 5A made paper wind turbines with 2, 3, 4, 6 and 8 blades. They put each one 1 m in front of the same fan and counted the turns in 30 seconds (three tries each).

1 **Complete the table: work out the mean (average) of the three tries.** [5]

Blades	Try 1	Try 2	Try 3	Mean turns
2	38	40	42	
3	54	57	57	
4	62	60	61	
6	52	55	55	
8	45	43	44	

2 **Draw a bar chart of the mean turns.** [3]



3 **Describe the pattern. Which number of blades worked best?** [2]

.....

.....

4 **Why did they do three tries and find the mean?** [1]

.....

.....

Data: example results for practice.

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REAL-WORLD CASE STUDY · C10 · ENERGY RESOURCES

Bangladesh's energy future

★★★
Grade 5

Name: **Class:** **Date:**

Learning goals

- Use data to evaluate a big decision.
- Consider different viewpoints.

Science · Geography · Economics · Citizenship

Bangladesh needs much more electricity for its homes, factories and growing cities. In 2023 it made over 85% of its electricity from fossil fuels: mostly natural gas, plus oil and a fast-growing amount of coal. Its own gas fields are running down, so it must buy more fuel from other countries, which is expensive.

At the same time, Bangladesh is one of the countries hit hardest by climate change. Many people say the country should build much more **solar** (on rooftops, over ponds and on unused land) and **wind** power along the coast, while saving energy. Others worry that solar needs a lot of land in a crowded country and does not work at night without batteries.

Based on Power Grid Company of Bangladesh data (2023) and public debate about Bangladesh's energy plans.

1 **Why must Bangladesh buy more fuel from other countries?** [2]

2 **Complete the viewpoints.** [3]

Person	What might they want? Why?
a coastal farmer	
a factory owner	
a city child with asthma	

3 **Write a short plan (3 actions) for green energy in Bangladesh. Use evidence.** [4]

Show what you know

★★★ Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the words.**

F	L	G	B	S	W	Y	G	S	B
F	S	O	R	I	P	K	P	C	D
T	M	O	T	E	O	W	W	L	W
R	B	N	L	C	E	G	N	H	I
D	W	M	M	A	O	N	A	F	N
I	C	E	I	D	R	A	I	S	D
C	L	I	M	A	T	E	L	J	B
G	L	V	T	C	U	M	L	J	P
A	U	H	Y	D	R	O	F	M	N
S	P	R	S	F	I	O	V	D	K

biogas climate coal gas green hydro solar wind

2 **True or false?** [3]

Statement	True	False	If false, correct it
1. Natural gas is a renewable resource.			
2. Most of Bangladesh's electricity in 2023 came from natural gas.			
3. Nuclear power releases a lot of carbon dioxide.			

3 **About what share of the world's electricity came from fossil fuels in 2024?** [1]

◯ about 20% ◯ about 59%

◯ about 90%

4 **Challenge. Your school wants to cut its electricity bill and its carbon footprint. Suggest three changes and explain each.** [3]

.....

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
sort renewable and non-renewable resources	○	○	○
read energy data and charts	○	○	○
explain why fossil fuels are a global issue	○	○	○
compare energy choices	○	○	○
plan green energy actions	○	○	○